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Paleomagnetic evidence for Cenozoic block rotations in the Tadjik depression (Central Asia)

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Abstract. This paper presents results of a paleomagnetic study of Oligo-Miocene red beds of the Tadjik depression in Central Asia. We sampled about 530 cores at 69 sites and six localities across the depression and along the western border of the Pamirs. Samples were thermally demagnetized and high-temperature components appear to predate folding of upper tertiary age. Throughout the depression, paleomagnetic inclinations are consistent with those observed on the stable Turan platform, at the western margin of the depression. However, they are shallower by about 30° than the inclination predicted from the reference apparent polar wander path. This appears to indicate a 23° difference in latitude, which is incompatible with paleogeographic reconstructions for the Tertiary. A sound interpretation of this anomaly would require a better-constrained Tertiary paleomagnetic reference for Asia. Inside the Tadjik depression, paleomagnetic declinations are all significantly rotated, counterclockwise with respect to those measured on the Turan platform. The eastern part of the depression is a domain of large rotation (52°±13° to 46°±15°), whereas smaller amounts of rotation have occurred in the western part (27°±14° to 14°±15°). The similarity between Tertiary and Cretaceous data available for the area shows that rotations have occurred since the Miocene. Little or no paleomagnetic rotations are observed in the ranges bordering the northern and western parts of the depression. Paleomagnetic and structural data suggest that block rotations in the Tadjik depression are associated with indentation of the Pamirs into stable Asia. At a larger scale, observed rotations are compatible with a model of regional sinistral wrenching, along a strip running from the Gulf of Oman to Lake Baikal.

Introduction

Central Asia provides a remarkable example of large-scale active intracontinental deformation. Over the last 55 m.y. [*Patriat and Achache*, 1984], the regional tectonics appear to have resulted from indentation of Asia by India [*Argand*, 1924; *Molnar and Tapponnier*, 1975], with a bulk N-S shortening of about 2000 km [*Achache and Courtillot*, 1984; *Savostin et al.*, 1986]. Among the various kinematic models [e.g., *Dewey and Burke*, 1973; *Achache and Courtillot*, 1984; *Houseman and England*, 1986, 1993; *Cobbold and Davy*, 1988; *Davy and Cobbold*, 1988; *Dewey et al.*, 1989; *Avouac and Tapponnier*, 1992; *Le Pichon et al.*, 1992] a key point, but a controversial one, is the partitioning between crustal thickening and strike-slip motions along major faults. Furthermore, the relative amount of convergence accommodated by eastward escape of large continental blocks is still debated [*Tapponnier et al.*, 1986; *England and Molnar*, 1990; *Houseman and England*, 1993].
The existence, the nature, the regional distribution, and the amount of block rotation about vertical axes are critical features for constraining the way bulk shortening has been accommodated throughout Asia [see *England and Molnar*, 1990]. From an analysis of the regional fault pattern (Figure 1), *Cobbold and Davy* [1988] inferred block rotations, clockwise in the eastern Himalayan syntaxis, counterclockwise in the western syntaxis, and along a wide strip of sinistral wrenching running from the Gulf of Oman to Lake Baikal. Recent paleomagnetic studies have consistently revealed clockwise rotations of up to 40° in Cretaceous rocks from eastern Tibet and s outhwest China [*Otofuji et al.*, 1990; *Huang and Opdike*, 1992; *Funahara et al.*, 1993], and counterclockwise rotations of up to 20° in Cretaceous and Paleogene rocks of intermontane basins of the Tien-Shan [*Thomas et al.*, 1993; *Bazhenov*, 1993]. The Tadjik depression, which lies South of the Tien-Shan and West of the Pamirs (Figure 2), is part of the Oman-Baikal sinistral wrench zone [*Cobbold and Davy*, 1988]. A few earlier paleomagnetic studies [*Abdullaev and Rzhevsky*, 1973; *Pozzi and Feinberg*, 1991], as well as field observations [*Thomas et al.*, 1994], have provided evidence for counterclockwise rotations in the Tadjik depression. We have undertaken a paleomagnetic sampling of Cenozoic and Cretaceous red beds in the depression and adjacent ranges in order to provide better constraints on the kinematic history of the area. This paper describes the paleomagnetic results for Tertiary samples. Results for the Cretaceous are described in detail by *Bazhenov et al.* [1994].

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Geological Setting

Tectonics

At regional scale, the Tadjik depression appears as an intermontane pushed-down basin, bounded by crustal scale fault

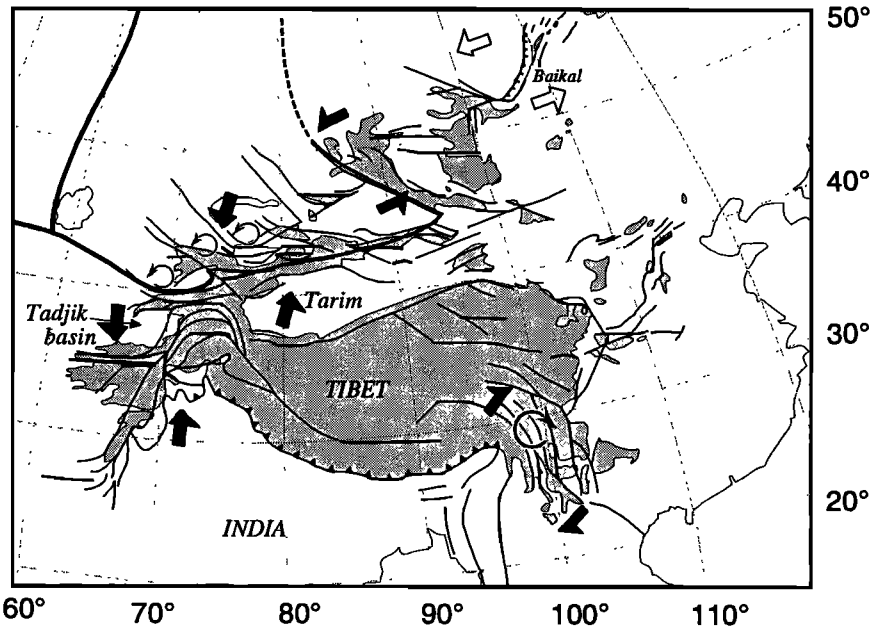
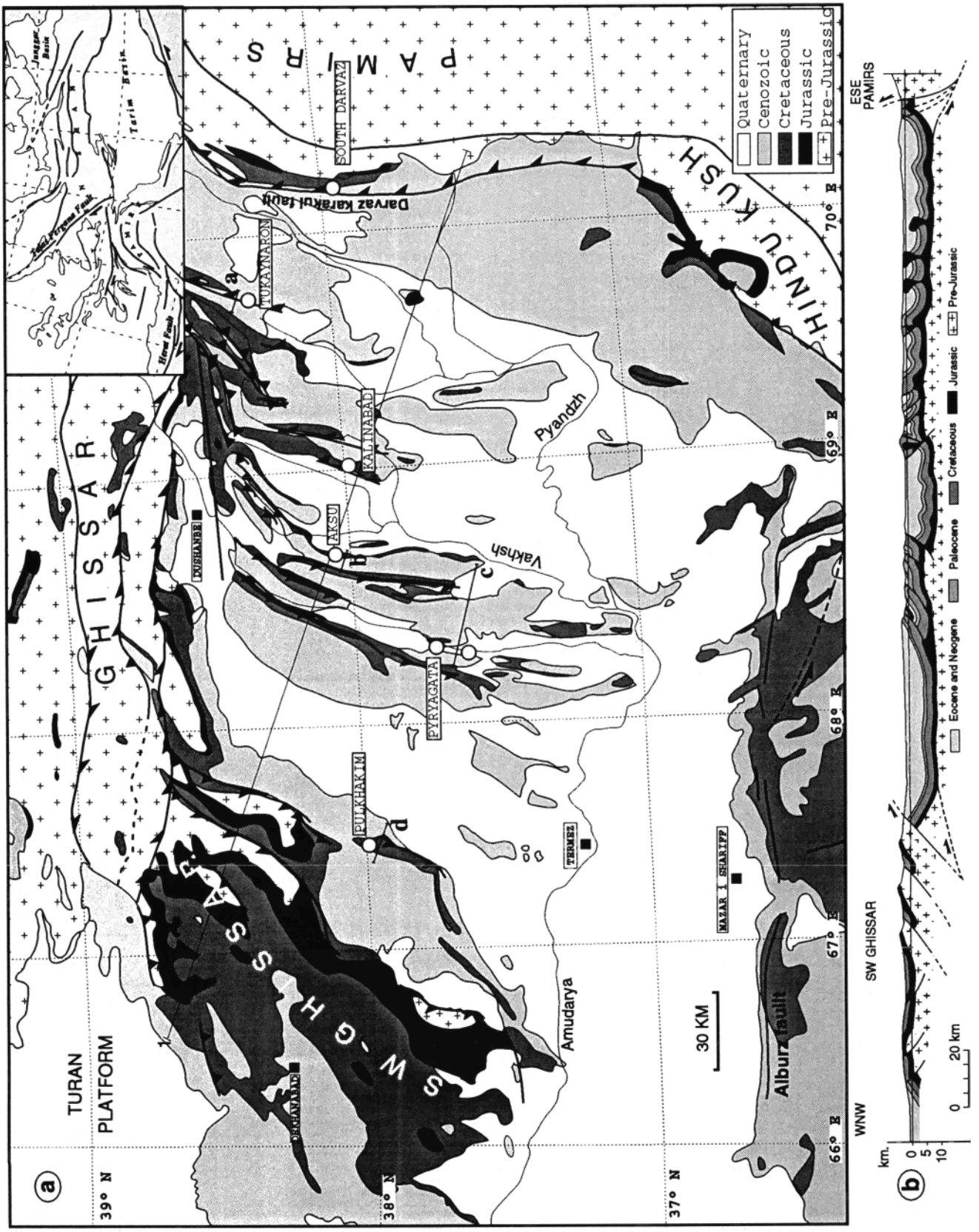


Figure 1. Cenozoic deformation in Central Asia [after *Cobbold and Davy*, 1988]. Map shows traces of major faults (thin lines), Paleozoic sutures (thick lines), areas with elevation greater than 2 km (stippled) and inferred Cenozoic displacement pattern. Pairs of arrows indicate horizontal displacement gradients (solid for contractional, open for extensional). solid half arrows indicate dominant wrenching in Altai and eastern Tibet; asymmetric solid arrows, convergent wrenching in Hindu-Kush and Tien-Shan; asymmetric open arrows, divergent wrenching in Baïkal. Round arrows indicate rotations relative to stable Asia.

zones with thrust and wrench components (Figure 2). In the east, the Darvaz-Karakul fault zone marks the boundary between the Pamirs and the depression. The relative motion of the Pamir wedge with respect to Asia is currently in a N-S to NNW-SSE direction [*Minster and Jordan*, 1978; *De Mets et al.*, 1990]. Seismological and geological data [*Hamburger et al.*, 1992] further suggest that the northeastern part of the Tadjik depression has been subducted under the Pamirs, a substantial part of the displacements being accommodated by westward thrusting and sinistral wrenching along the Darvaz-Karakul fault zone (Figure 2a) [*Peive et al.*, 1964; *Burtman and Molnar*, 1992]. The Ghissar range of the southwestern Tien-Shan forms the northern margin of the depression. It consists of Hercynian basement, reactivated during the Tertiary [*Burtman*, 1975] and is considered a major and still active right-lateral wrench zone [*Trifonov*, 1978; *Thomas et al.*, 1994], with a component of southward thrusting [*Ministry of Geology of the USSR and Tadjik Academy of Sciences*, 1984]. To the west, the SW Ghissar ranges separate the Tadjik depression from the Turan platform. Here eastward thrusting is dominant (Figure 2b). Local thrusting of basement rocks over the Tertiary sediments of the depression suggests thick-skin tectonics. South of the Amudarya river, the east-west striking Alburz fault separates Quaternary deposits of the Tadjik basin from Tertiary and Mesozoic sediments of Northern Afghanistan. Seismological data show that this fault is currently active [*Heuckroth and Karim;* 1973, *Shareq*, 1981]. From geological maps [*Ministry of Geology of the USSR and Tadjik Academy of Sciences*, 1984] and unpublished subsurface data, there is a component of northward directed thrusting. The occurrence of en echelon folds located in the vicinity of the eastern termination of the Alburz fault [*Ministry of Geology of the USSR and Tadjik Academy of Sciences*, 1984] suggests that thrusting is combined with a component of sinistral wrenching. Farther south, the Herat fault is considered to be a Mesozoic suture, reactivated as a dextral strike-slip fault during the India-Eurasia collision [*Tapponnier et al.*, 1981].

The Tadjik depression itself is a succession of alternating basins and ranges, striking nearly north-south (Figure 2). Basins are generally bounded by thrusts and asymmetric faulted folds with oblique-slip components [*Thomas et al.*, 1994]. Thrusting is partly thin-skinned, above a décollement level of Upper Jurassic evaporites [*Leith and Simpson*, 1986; *Thomas et al.*, 1994] (Figure 2). Thrusts are of westward vergence in the eastern part of the depression (synthetic to the Darvaz-Karakul fault, eastern depression boundary) and of eastward vergence in the western part (synthetic to thrusting along the SW Ghissar ranges) (Figure 2). Deformation in the depression and uplift of surrounding ranges mainly occurred during late Miocene to Quaternary times, as indicated by an increasing supply of continental sediments during this period. Shallow seismic activity is widespread in the eastern part of the depression, especially in the strongly deformed area between the Pamirs and the South Tien Shan (Peter the First range) [*Leith and Simpson*, 1986; *Hamburger et al.*, 1992]. Quaternary faulting is also common throughout the depression [*Trifonov*, 1978].

Figure 2. (a) Simplified geological map of the Tadjik depression. Open circles are paleomagnetic localities. Lines 1, a, b, c, and d correspond to cross sections of Figures 2b and 3a to 3d, respectively. Oblique triangles on faults point in direction of underthrusting. Inset map shows main tectonic features of Central Asia (shaded areas are Mesozoic and older units). (b) Cross section through the Tadjik depression. Data are from *Zakharov* [1962], *Ministry of Geology of the USSR and the Tadjik Academy of Sciences* [1989], and our observations.


a
TURAN
PLATFORM
G H I S S A R
P A M I R S
H I N D U K U S H
S W - G H I S S A R - A L A I
Darvaz Karakul fault
SOUTH DARVAZ
TUKAYNARON
KALINABAD
AKSU
PUSHANBE
PYRYAGATA
PUL-KHAKIM
TERMEZ
KAZAR 1 SHARTEP
Pyandzh
Vakhsh
Amudarya
Alburz fault
30 KM
39° N
38° N
37° N
66° E
67° E
68° E
69° E
70° E
Quaternary
Cenozoic
Cretaceous
Jurassic
Pre-Jurassic
WNW
SW GHISSAR
ESE
PAMIRS
Eocene and Neogene
Paleocene
Cretaceous
Jurassic
Pre-Jurassic
km.
0
5
10
20 km
b

Sedimentology

Sediments in the Tadjik depression and SW Ghissar area range in age from Lower Jurassic to Quaternary. The total thickness of the cover can reach 10 km. Neogene and Quaternary deposits constitute roughly half of this amount. They are particulary thick in the center of the depression and toward its eastern and western margins (Figure 2b).

The Jurassic sequence consists of Hettangian to Kimmeridgian limestones, overlain by late Jurassic evaporites which provide a major décollement level. The thickness reaches up to 2.5 km in the SW Ghissar ranges. Overlying Cretaceous deposits are about 2 km thick and change gradually from continental in the lower cretaceous to shallow marine in the upper cretaceous.

Paleogene rocks are generally conformable on the Upper Cretaceous; but slight angular unconformities occur locally. The whole Paleogene sequence is commonly several hundred meters thick. However, considerable lateral changes in facies and thickness are observed. A change from shallow marine to continental environment occurs during the Oligocene.

The overlying Baljuan Series, of Miocene (and perhaps upper Oligocene) age, consist of intercalated red sandstones and conglomerates. They are overlain by Pliocene to Quaternary alluvial sandstones and conglomerates. Our paleomagnetic study was mainly performed on Oligo-Miocene red beds, especially sandstones and siltstones.

Paleomagnetic Analysis

We collected 530 cores at five localities across the depression, between the Pamirs and the southwest Ghissar ranges (Figure 2a). Each locality consisted of several sites where seven to nine cores were collected in two or three adjacent beds. Both solar and magnetic compasses were used for orientation. Sites were located precisely with a portable satellite positioner (Global Positioning System).

Remanent magnetization was measured with a cryogenic magnetometer (LETI). We isolated characteristic directions of remanent magnetism after progressive thermal demagnetization. Alternating field demagnetization was found to be less efficient.The bulk susceptibility was measured at each step during demagnetization. Characteristic paleomagnetic directions were determined by principal component analysis. A complete reinterpretation of data from the outskirts of the Pamirs (South Darvaz locality), previously published by *Bazhenov and Burtman* [1986], is also included in this study.

Lithology and Magnetic Mineralogy

Natural remanent magnetization (NRM) was found to range in intensity between 10^{-3} and 10^{-2} A/m for the overall collection. All samples that were found to carry stable components of magnetization during demagnetization presented unblocking temperatures higher than 600°C. The bulk susceptibility did not reveal significant changes during thermal demagnetization, showing that no chemical transformation of the magnetic carriers occurred during heating. Isothermal remanent magnetization (IRM) was given to representative samples from each locality (Figure 3). Magnetization never reached saturation for an applied field of 1.2 T. IRM acquisition curves for Tukaynaron locality (Figure 3a) indicate magnetic minerals with high coercivity, probably hematite. For the other localities, curves show two stages. First, for fields smaller than 0.2 T, IRM intensity increases rapidly; then intensity increases much more slowly but

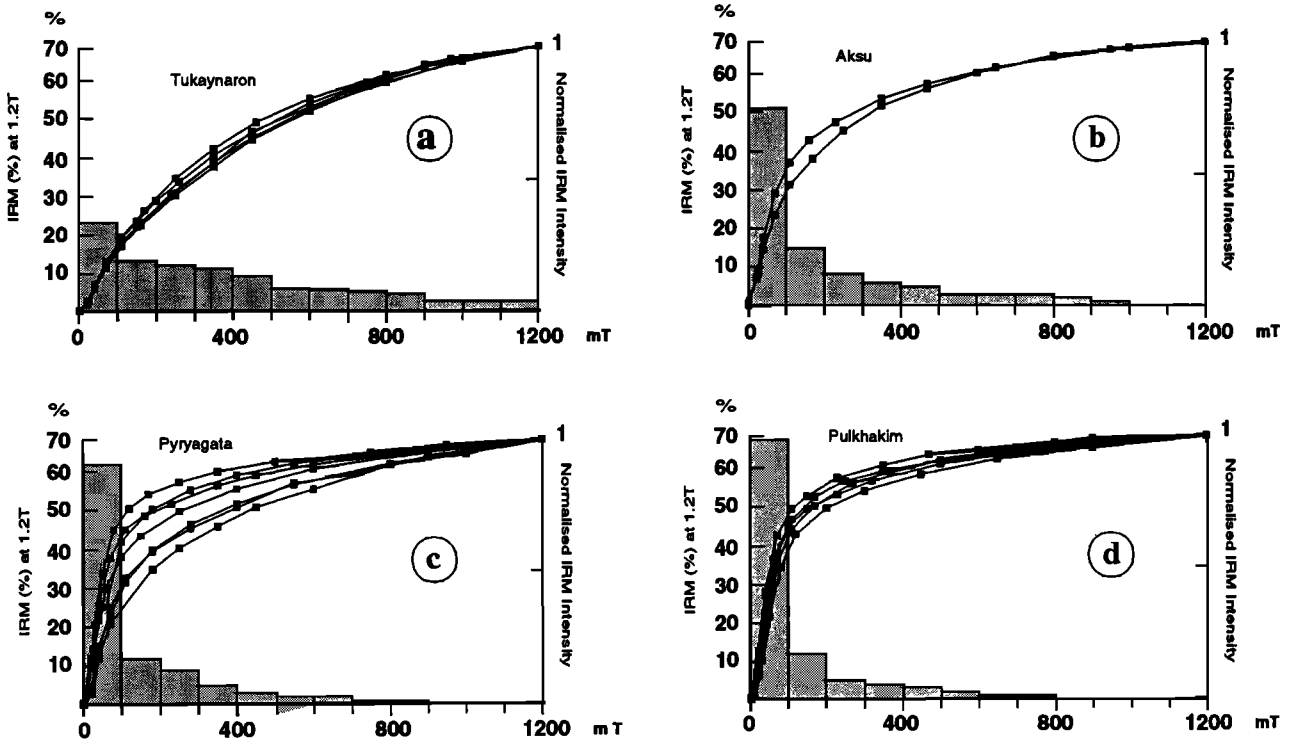


Figure 3. Normalized isothermal remanent magnetization curves and coercivity spectra (in grey) for representative samples from various localities. Scale on left-hand side of diagrams shows percent of IRM at 1.2 T. Hematite is dominant for Tukaynaron locality, whereas both hematite and magnetite are present for other localities. (a) Tukaynaton, (b) Aksu, (c) Pyryagata, (d) Pulkhakim.

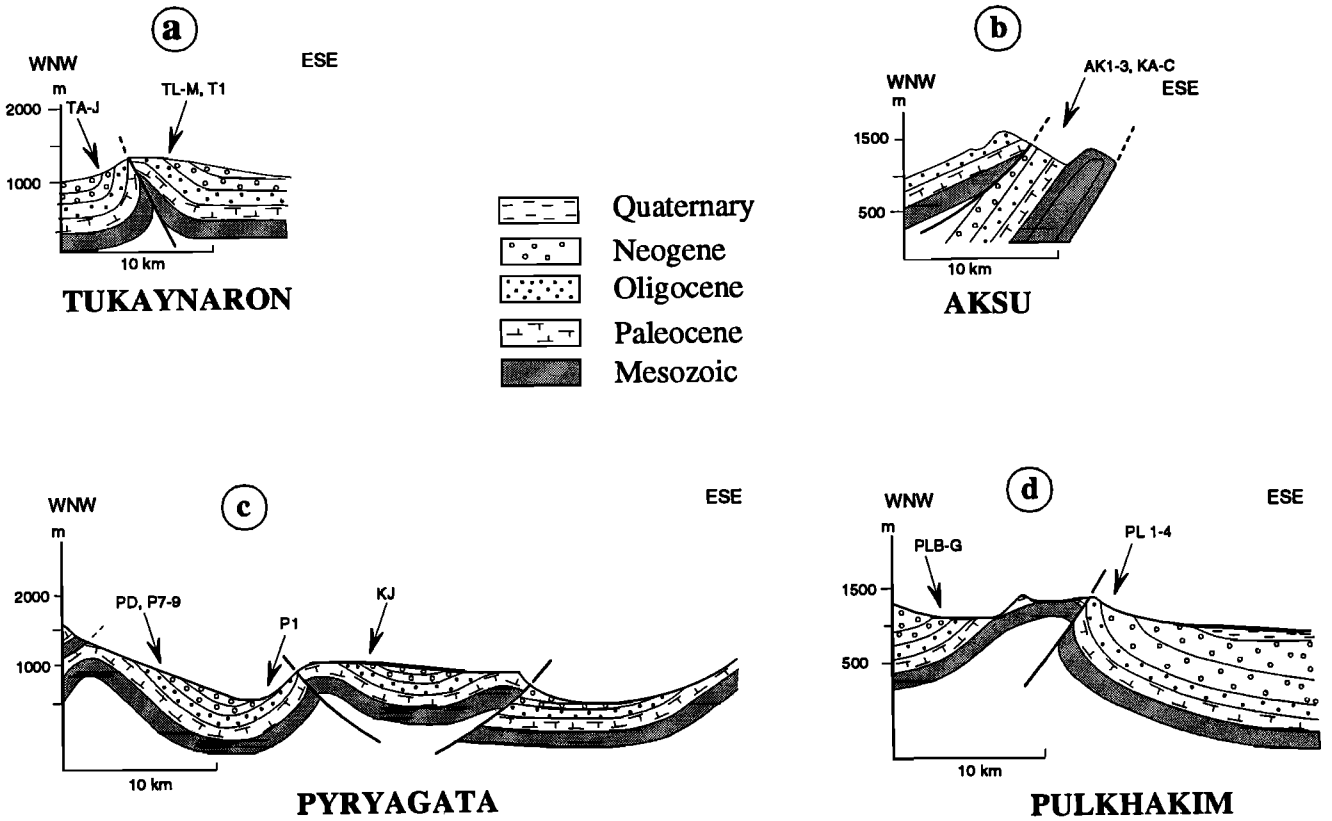


Figure 4. Schematic cross sections located on Figure 2 for four out of six localities. Arrows indicate locations of sampling sites. (a) Tukaynaton, (b) Aksu, (c) Pyryagata, (d) Pulkhakim.

does not reach saturation. We infer two magnetic phases, magnetite and hematite. From both unblocking temperatures and IRM curves, the high-temperature component of our samples is carried mainly by hematite, even for samples where the proportion of magnetite is significant. For Tukaynaron locality, where hematite is clearly dominant on IRM curves, samples are fine-grained reddish sandstones. For the other localities, samples are more brownish to pink in color and the grain size is larger. Representative polished sections from all localities, observed under a reflected-light microscope, show that samples consist of clast-supported sandstones and siltstones essentially made of quartz. Clasts are angular and their size varies from about 60 mm in sandstones to less than 10 mm in siltstones. Spaces between quartz grains are filled with a reddish matrix, at least partly composed of hematite pigment which probably carries most of the NRM. Large specular hematite grains are locally observed.

Paleomagnetic Results

Tukaynaron. The Tukaynaron locality is located in the Northeastern part of the Tadjik depression, at the closure of an anticline. On both limbs of the fold (Figure 4a), we sampled 18 sites in fine-grained reddish sandstones of upper Oligocene to lower Miocene age. Most samples yielded two components of magnetization after thermal demagnetization (Figures 5a and 5b). The low-temperature component, removed around 350°C to 500°C, is close to the current dipole field direction. The high-temperature component has an unblocking temperature between 650°C and 680°C. Both polarities of the Earth's magnetic field are observed. As the axis of the studied fold plunges 42° toward 192°, we corrected data both for tilt and for fold axis orientation (unfolded, Table 1). Both fold test [*Mc Elhinny*, 1964] and reversal test are positive at the 99% level (Figure 5c).

Kalinabad. The Kalinabad locality, a syncline is exposed along the Vakhsh river. Ten sites were sampled on both limbs. Unfortunately, only two sites provided results with clear characteristic components (Figure 6a). Other sites showed erratic or unstable behavior during thermal demagnetization. This we attribute to the magnetic mineralogy and to recent weathering of sediments. Indeed, on polished sections, the rock is strongly altered. Intergranular spaces are filled with a grey matrix replacing the original pink matrix which contained the hematite. In tilt-corrected coordinates, the two sites revealed antipodal paleomagnetic directions (Figure 6b and Table 2). However, the number of sites is too small for statistics and field tests.

Aksu. At the Aksu locality, in the center of the depression, six sites were sampled on one limb of an anticline (Figure 4b). Three sites (AK1 to AK3) were collected in lower Miocene fine-grained reddish sandstones; the three other sites (KA to KC), a few kilometers to the south, in pale brownish sandstones. Thermal demagnetization (Figures 7a and 7b) revealed two components: a low-temperature component with a direction close to the present field and an unblocking temperature around 300°C to 450°C; and a high-temperature component, usually very stable and completely removed around 680°C. These two components can have antipodal directions (e.g., Figure 7a), and the high-temperature component is mostly of reverse polarity. Unfortunately, bedding dips do not vary enough to perform a fold test (Figure 7c and Table 3). Structural observations show that sites AK1 to AK3 are located on a fold with an axis plunging 22° toward N7°. Cretaceous samples were collected on both limbs of

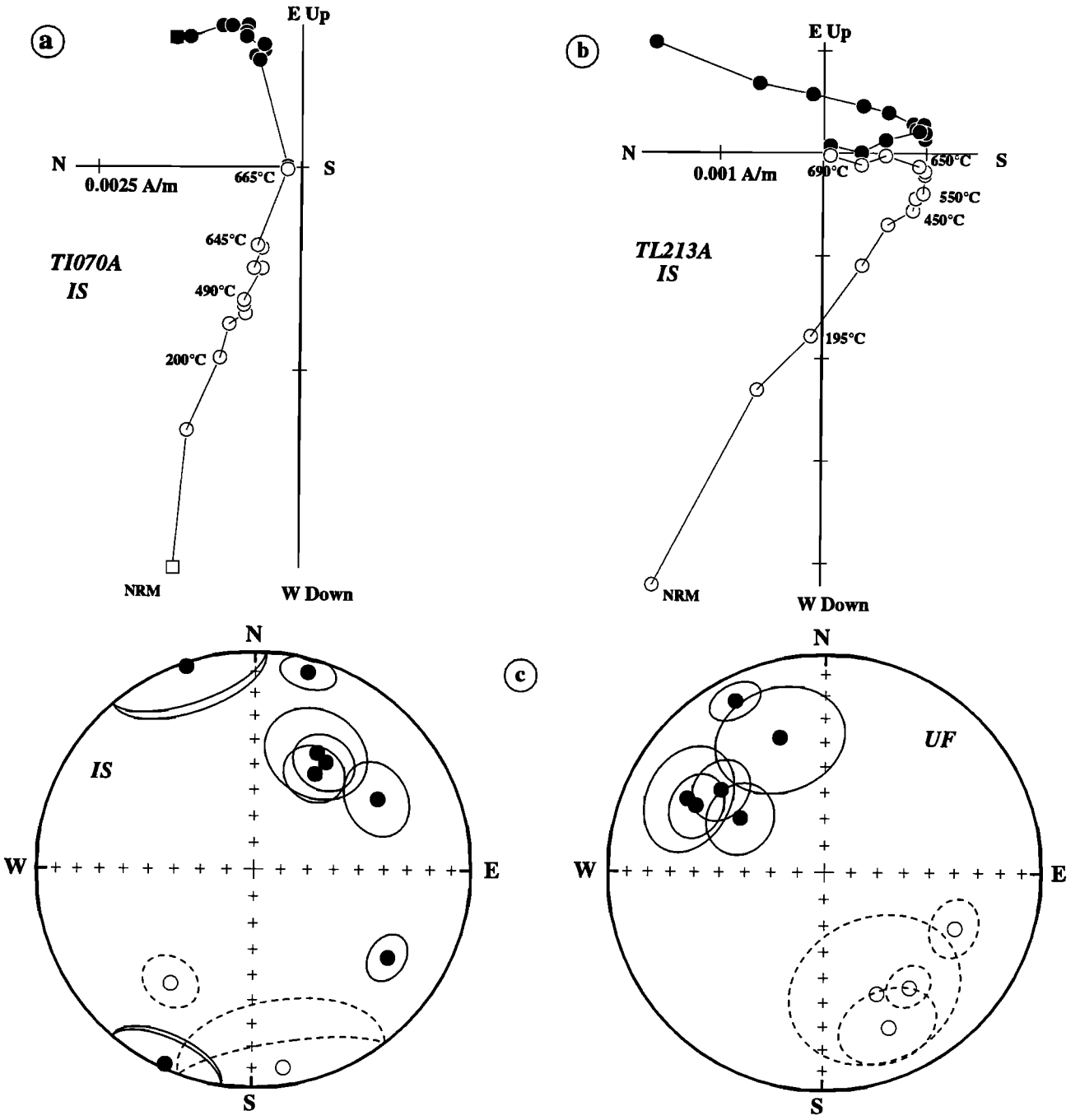


Figure 5. Paleomagnetic data for Tukaynaron locality. (a) and (b) Representative demagnetization curves. Solid and open symbols are in horizontal and vertical plane respectively. (c) Equal-area projections of mean site directions (IS, in situ coodinates; UF, unfolded coordinates, see text of Table 1). Solid and open circles are projections in the lower and upper hemisphere, respectively.

this fold. Corresponding paleomagnetic directions showed better clustering after unfolding than after simple tilt correction [*Bazhenov et al.*, 1994]. In situ directions for sites AK1 to AK3 were therefore unfolded. Site KB was rejected for the calculation of the mean direction because of its strongly deviant direction in tilt-corrected coordinates with respect to the other sites.

Pyryagata. At the Pyryagata locality, in the south central part of the depression, 19 sites were sampled in lower Miocene rocks on three limbs of a syncline-anticline pair (Figure 4c). As for Kalinabad locality, many samples proved unstable during thermal demagnetization. These samples were often coarse-grained with a pale-brownish color, probably induced by weathering. Nevertheless, six sites provided reliable results. Two types of behavior were observed: one set of samples showed a unique component (Figure 8a) which is very stable, with unblocking temperatures distributed between 100°C and 680°C. For the other samples (Figure 8b), a low-temperature component was always removed after 550°C and a high-temperature component was usually removed after 650°C; this latter component is not so well defined as for the first set of samples. Despite the relatively small

Table 1. Paleomagnetic Data for Tukaynaron Locality

Site	Latitude, °N	Longitude, °E	Bedding	n/N	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I	Unfolded D	Unfolded I	k	α95
TA	38.338	69.640	186/66	5/8	032	47	324	34	297	34	48	11
TB	38.338	69.640	184/71	7/8	015	07	001	12	332	12	36	8
TC	38.338	69.640	183/74	7/8	028	39	327	29	298	29	12	18
TD	38.338	69.640	179/76	8/8	204	01	186	-24	157	-24	14	16
TE	38.338	69.640	181/79	7/7	216	-36	145	-35	113	-35	41	10
TI	39.333	69.633	188/73	8/8	060	35	332	52	302	52	18	13
TJ	39.333	69.633	184/62	7/8	033	41	332	39	308	39	30	11
TL	39.333	69.633	116/40	4/7	171	-09	158	-39	156	-39	10	30
TM	39.333	69.633	89/35	7/7	341	02	337	35	341	35	8	22
T1	38.300	69.648	45/63	6/9	123	27	122	-35	143	-35	70	8

	N	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I	Unfolded D	Unfolded I	k	α95
	10	17	18					4	25
				335	30			19	11
						317	33.5	19	11

Top part of table gives site label, site location, bedding attitude, number of samples used for statistics (n) versus number of samples measured (N), mean declination (D) and mean inclination (I) for high-temperature components in in situ, tilt-corrected, and unfolded coordinates, Fisher statistics (k for dispersion parameter, α_{95} for radius of circle of confidence at 95% probability level), and approximate stratigraphic age. Data were tilt-corrected by rotating to horizontal about local strike; unfolded, by first rotating about an orthogonal to the fold axis and then rotating to horizontal about local strike. Bottom part of table gives number of data used for calculation (N), mean declination (D) and inclination (I) for each locality, and Fisher statistics for high-temperature components. Ages are upper Oligocene for sites TA to TJ and lower Miocene for sites TM to T1.

number of sites, both polarities are observed (Figure 8c and Table 4) and a fold test is positive at the 99% level [*Mc Elhinny*, 1964].

Pulkhakim. Pulkhakim locality is on the western margin of the depression. We sampled 12 sites on both limbs of a faulted box fold (Figure 4d), within reddish to brownish sandstones of lower Miocene age. During thermal demagnetization, most samples showed a well-defined characteristic component. As for the Pyryagata locality, two types of behavior are observed during thermal demagnetization. Figure 9a shows an example with a single characteristic component converging towards the origin and with an unblocking temperature around 680°C. Figure 9b shows more complex behavior. Up to 600°C, no clear component can be isolated. This is probably due to simultaneous removal of two components carried by magnetic minerals with overlapping spectra of unblocking temperatures. Between 620°C and 680°C, a high-temperature component can be isolated. At the scale of the locality (Figure 9c and Table 5), this component presents both polarities of the Earth magnetic field. A reversal test is positive; so is a fold test at the 99% level [*Mc Elhinny*, 1964].

South Darvaz. South Darvaz locality is on the eastern border of the Tadjik depression, along the Pamir boundary. We sampled siltsones and grey or brownish-red sandstones along three sections through steeply dipping beds of Eocene to early Miocene age. At section S1, 19 samples were taken through Eocene to lower Miocene beds; at the other two, S2, 9 samples and S3, 12 samples, Oligocene to lower Miocene rocks were studied. This collection was previously studied by *Bazhenov and Burtman* [1986] with only partial demagnetization up to 400°C. Demagnetization was completed at Rennes in 1992. Most demagnetization curves show a single-vector component from 100°-200°C to 680°C (Figure 10a). Despite a well-defined component for most samples, paleomagnetic directions are rather scattered (Figure 10b and Table 6). Both polarities are observed (Figure 10b) and the reversal test is positive. No fold test could be performed, because bedding attitudes are almost uniform. The way of sampling for this locality differs from that at the others: the number of sites (sections) is smaller while the number of samples per sites is larger. Although we calculated the locality mean direction both at the sample and the site level (Table 6), we preferreed to use that on sample level since each hand sample represents an independent, though poorly defined, spot reading of the remanence.

Discussion on Paleomagnetic Data

Quality and Reliability of Data

At two localities (Tukaynaron, Pulkhakim), despite positive fold tests, data are still rather scattered at the between-site level after tilt correction (k=19 and k=16). Thus the observed scattering in tilt-corrected coordinates cannot be due to magnetization acquired during deformation. *Butler* [1992] suggested that the magnetization of red beds, similar to those from Tukaynaron and Pulkhakim, is a chemical remanent magnetization (CRM) acquired between 10^2 and 10^5 years after sedimentation. If the CRM is acquired before 10^3 years, then the between-site scattering can be explained by an acquisition of remanence faster than the time necessary to satisfactorily average the secular variation of the geomagnetic field. Nevertheless, at Tukaynaron and Pulkhakim localities, a substantial number of sites (10) have mean directions of overall good quality. We thus consider that the bulk mean directions for these localities are reliable.

For Kalinabad and Pyryagata localities, we rejected a large number of sites for which it was impossible to obtain well-defined high-temperature components. This probably reflects strong weathering of rocks from these outcrops; actually, we had great difficulty in finding red beds not too weathered and

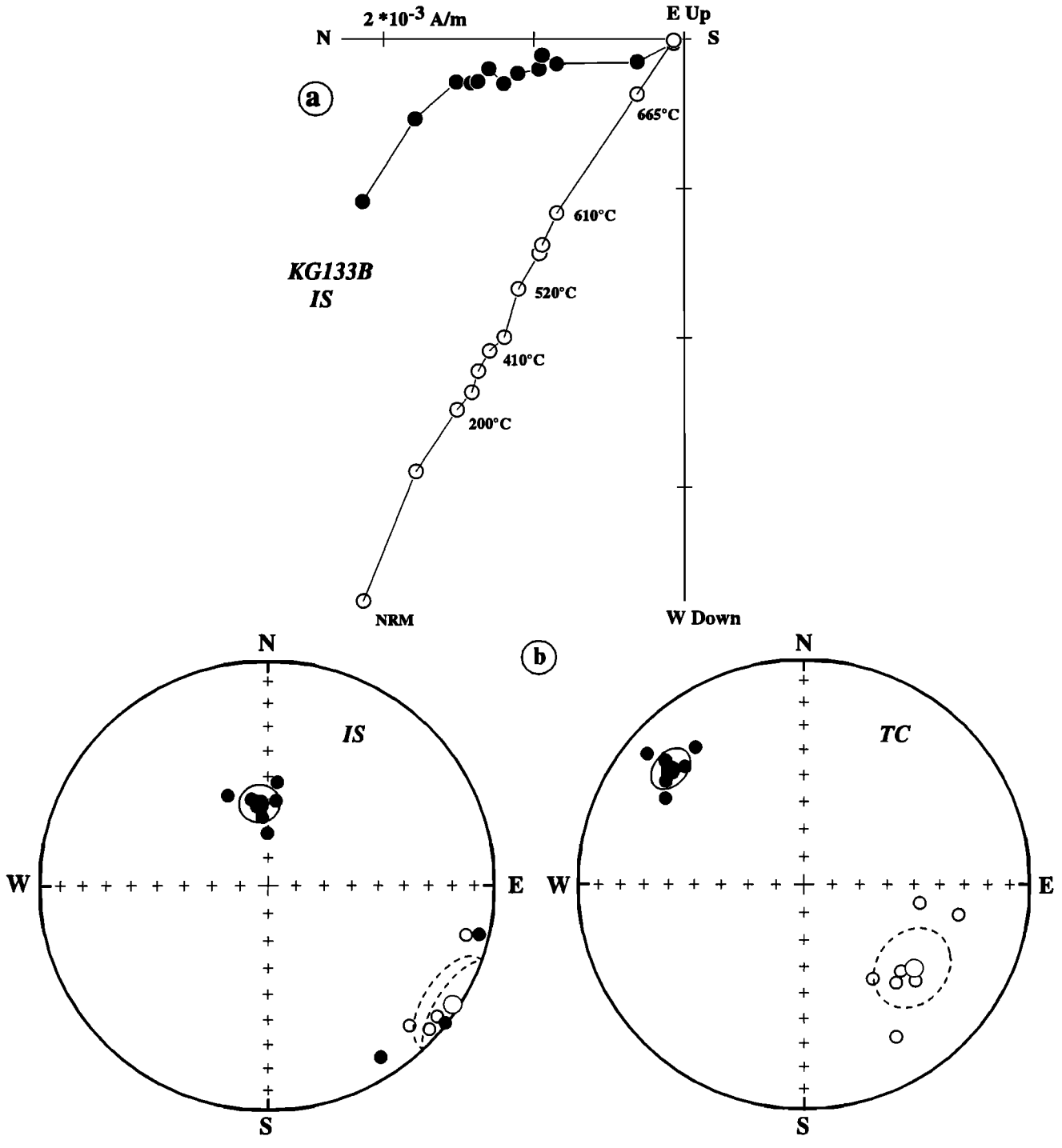


Figure 6. Paleomagnetic data for Kalinabad locality. Projections show both sample and site mean directions; TC is for tilt-corrected coordinates; other comments as on Figure 5.

therefore suitable for paleomagnetic study in these areas. Prolonged CRM acquisition, over many more than 10^5 years after sedimentation, will very likely encompass polarity changes and may lead to large scatter of the paleomagnetic data if separation of the components is not optimal [*Bazhenov and Shipunov*, 1991]. However, for Pyryagata locality, the six remaining sites show antipodal directions and a good clustering after tilt correction. We therefore consider the mean direction to be reliable. At Kalinabad locality, only two sites yielded a reliable mean direction. Given that only few Tertiary paleomagnetic data exist for the Tadjik depression, we have not rejected this locality, keeping in mind that corresponding data must be interpreted with caution.

In general, fold tests are positive. For two localities (Aksu and South Darvaz), fold tests could not be done. However, positive fold tests were already obtained on Cretaceous rocks from the same regional structures (Table 7) [*Bazhenov et al.*, 1994], but with a steeper mean direction. The good consistency of tilt-corrected results between all localities suggests that magnetization was acquired before folding.

Table 2. Paleomagnetic Data for Kalinabad Locality

Site	Latitude, °N	Longitude, °E	Bedding	n/N	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I	k	α_{95}
KG	37.967	69.011	189/55	6/8	354	60	311	23	76	7
K1	37.964	68.996	26/37	7/7	125	-3	128	-39	20	14

	N	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I	k	α_{95}
	13	320	34			5	20
				310	33	25	9

Due to small number of sites, statistics are calculated at sample level. Ages are lower Miocene. Other comments as for Table 1.

Paleomagnetic Reference

Discussion and interpretation of paleomagnetic data require a reliable reference, established for undeformed areas. Over the last 10 years, several authors have calculated apparent polar wander paths (APWPs) for Eurasia [*Irving and Irving*, 1982; *Westphal et al.*, 1986; *Besse and Courtillot*, 1991]. However, some paleomagnetic studies of Tertiary sediments from Asia [*Westphal et al.*, 1986; *Bazhenov and Burtman*, 1986; *Huang and Opdike*, 1992; *Thomas et al.*, 1993] have revealed major differences between observed directions and those expected from APWPs, especially for inclinations. Furthermore, most APWPs for the Cenozoic are based only on data from Western Europe, very few or none on data from Asia.

To obtain a more reliable regional reference, we sampled Tertiary rocks on the Turan platform adjacent to the Tadjik basin. We collected lower Miocene rocks on the western margin of the SW Ghissar ranges, near Dekhanabad city (Figure 2a). There, sediments of the platform are only slightly tilted. Cretaceous rocks were also sampled in the SW Ghissar ranges. For these rocks, *Chauvin et al.* [1992] and *Bazhenov et al.* [1994] show that there are no significant paleomagnetic differences in inclination and in declination between the measured Cretaceous direction and the direction predicted by the APWP of *Besse and Courtillot* [1991] at 100 Ma. Dekhanabad locality, which belongs to the same structural unit, can thus be considered as a stable area since the Late Cretaceous. We therefore consider the mean paleomagnetic direction of this locality as a local Tertiary reference for the Tadjik basin and adjacent areas. The mean pole for the locality is latitude 68°N, longitude 239°E [*Chauvin et al.*, 1992]. Compared with the APWP curve of *Besse and Courtillot* [1991] at 30 Ma (latitude: 81°N, longitude: 132°E, Table 7), there is no significant difference in declination; but the inclination at Dekhanabad is 30°±7° lower than expected. This anomaly will be discussed in the following section.

Inclination

In Figure 11a and Table 7, inclinations observed on Tertiary rocks within the Tadjik depression and along its margins are compared with the APWP of *Besse and Courtillot* [1991] and reference poles from *Van der Voo* [1990]. The most remarkable feature is the very good consistency between localities, including the reference locality of Dekhanabad. Inclinations range from 30°±12° to 35°±13°. This emphasizes that Tertiary inclinations are actually lower by about 30°±10° than those predicted by the APWP (≈60°). Furthermore, the inclination discrepancy varies very slightly for different reference data. Note that the APWP of *Besse and Courtillot* [1991] for the Cenozoic is almost stationary (see Figure 11a). This excludes any anomaly due to uncertain ages of sediments. Comparable inclination anomalies have already been described for Tertiary sediments of the Tien-Shan

Table 3. Paleomagnetic Data for Aksu Locality

Site	Latitude, °N	Longitude, °E	Bedding	n/N	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I	Unfolded D	Unfolded I	k	α_{95}
KA	37.961	68.549	170/24	6/8	182	-48	155	-48			36	11
KB*	37.961	68.549	194/26	4/8	034	29	18	35			19	21
KC	37.961	68.549	198/24	8/8	180	-56	155	-43			19	13
AK1	38.105	68.608	197/70	7/8	212	-36	165	-23	180	-23	20	15
AK2	38.105	68.608	197/75	7/8	219	-37	162	-26	179	-26	32	11
AK3	38.105	68.608	198/71	4/8	212	-55	143	-23	160	-23	14	25

	N	In Situ D	In Situ I	Tilt corrected D	Tilt corrected I	Unfolded D	Unfolded I	k	α_{95}
	5	22	48					25	16
				336	33			33	13
						347	33	26	15

Ages are lower Miocene. Other comments as for Table 1.
*Rejected site.

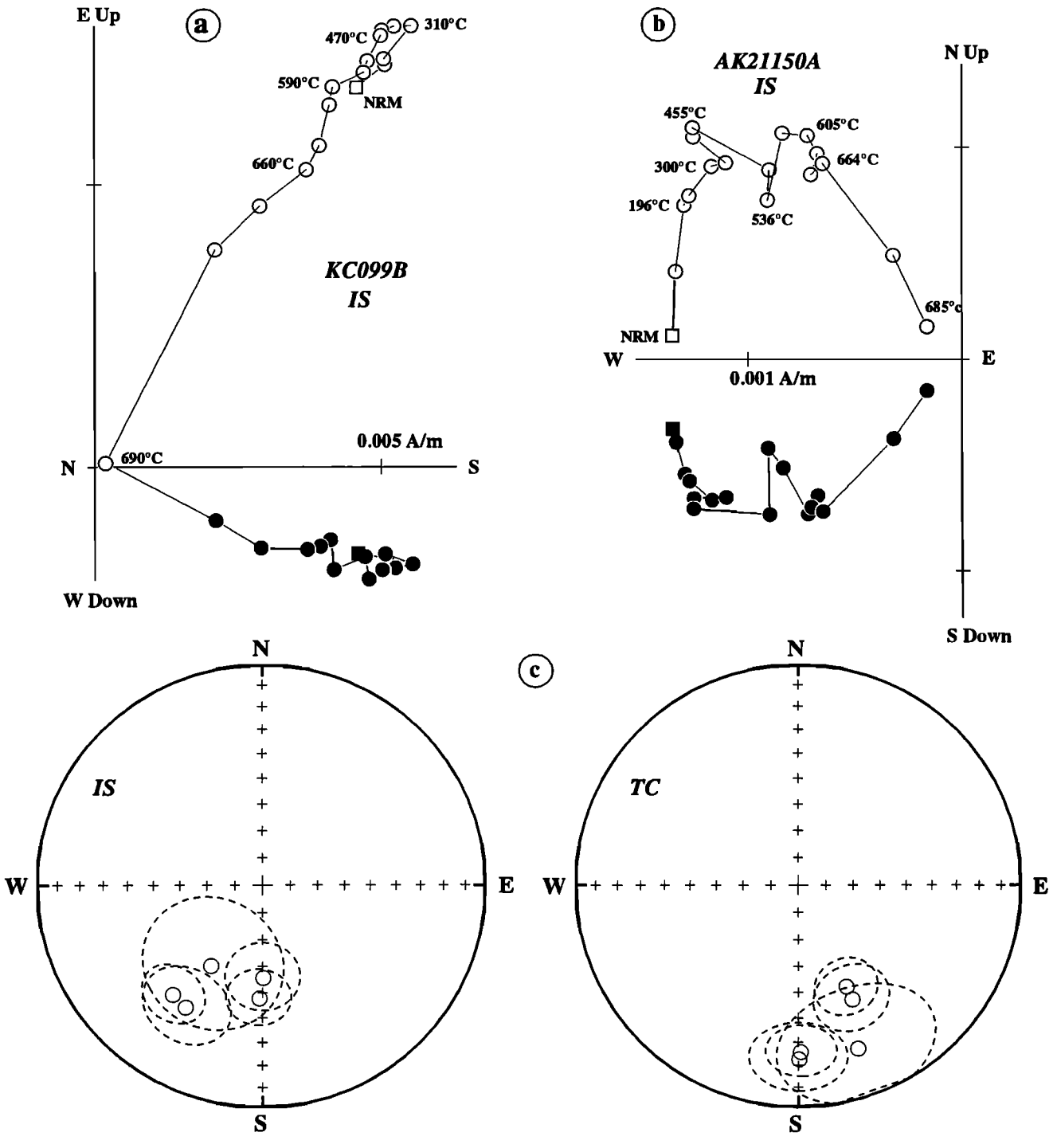


Figure 7. Paleomagnetic data for Asku locality. Same comments as for Figure 5.

range, adjacent to the Tadjik depression [*Thomas et al.*, 1993]. At the scale of Eurasia, inclination anomalies have also been reported elsewhere in the Tethyan belt [*Westphal et al.*, 1986; *Westphal*, 1993] and especially in the Aegean area [*Kissel and Laj*, 1988; *Beck and Schermer*, 1994].

The usual reason invoked for a difference between expected and measured inclinations is a displacement in latitude. If so, the observed inclination anomaly would require at least 2500 km of northward displacement of the Tadjik depression, whereas no more than a few hundreds of kilometers are readily accounted for by Cenozoic deformation north of the Tadjik depression, in the Tien-Shan ranges and in the Kazakhstan platform [*Avouac and Tapponnier*, 1992; *Avouac et al.*, 1993; *Thomas et al.*, 1993]. Moreover, the Late Cretaceous inclination is 30° steeper than the Tertiary inclination, which would further imply a 1500-km southward motion before the hypothetic Tertiary northward displacement. Such a history of motion is not supported by any tectonic or geodynamic data. Finally, it was noted in the previous section that the APWP of Eurasia for the Tertiary has been obtained from European data only. Thus this APWP cannot exclude the possibility of Cenozoic relative motion between Asia and Europe, which might be accommodated, for example, by

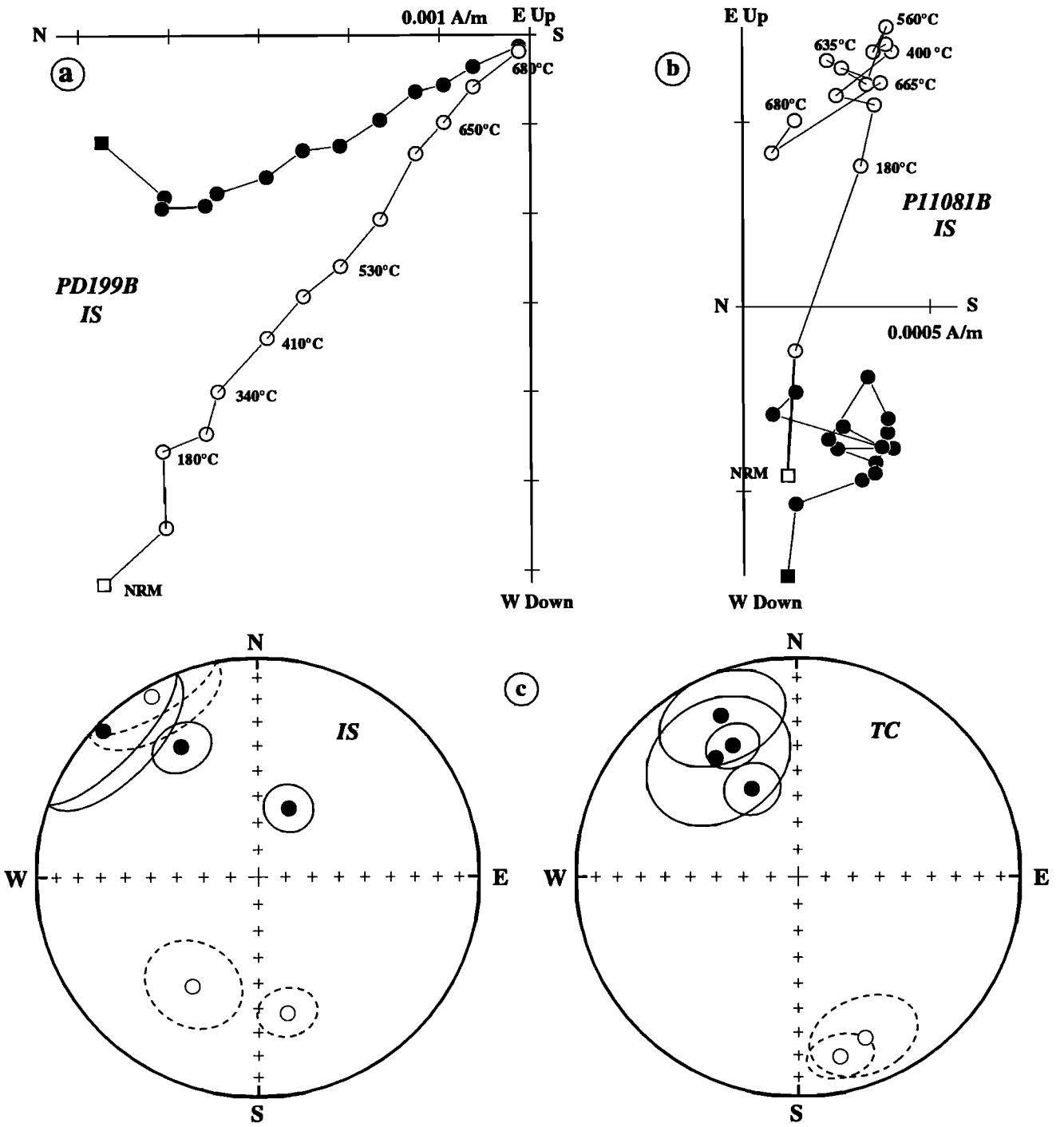


Figure 8. Paleomagnetic data for Pyryagata locality. Same comments as for Figure 5.

reactivation of Paleozoic sutures or orogenic areas (see Figure 1). However, no Tertiary tectonic features have been reported so far in the Siberian-Kazakhstan platform which could accommodate a relative northward displacement of several hundreds of kilometers between Europe and Asia. We thus consider that the Tadjik depression and the adjacent Tien-Shan range [*Thomas et al.*, 1993] did not suffer tectonic displacements, relative to Asia and Eurasia, large enough to create more than a few degrees of inclination anomaly.

Anomalous behavior of the geomagnetic field may induce errors in the reference data. *Schneider and Kent* [1990] showed that a positive axial quadrupole term (around 3% or 4% of the dipole term) might have been present throughout the Tertiary. However, this leads to a negligible difference in inclination (1.5° at 40° latitude) in comparison with a pure dipole. Recently, *Westphal* [1993] has suggested that, during the Eocene, the Earth's magnetic field was not that of an axial dipole but had an axis inclined at 18° to the rotation axis. This interpretation is based on data from the Eurasian Tethyan belt. We do not favor it, for several reasons. First, our sediments are mainly lower Miocene in age, which would imply an abnormal behavior of the geomagnetic field during at least 40 Ma. Second, the data selected by *Westphal* [1993] are not always convincing. In particular, ages are not all precisely constrained (see discussion

Table 4. Paleomagnetic Data for Pyryagata Locality

Site	Latitude, °N	Longitude, °E	Bedding	n/N	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I	k	α_{95}
P1	37.830	68.230	197/72	4/8	211	-42	158	-22	31	17
P7	37.507	68.031	214/45	8/8	24	62	334	34	37	09
P8	37.715	68.150	1/45	6/8	314	4	326	35	8	25
P9	37.715	68.150	6/45	6/7	330	-5	335	20	12	20
PD	37.531	68.029	197/75	7/8	330	32	333	53	30	12
KJ	38.105	68.608	198/71	6/8	168	-37	167	-17	49	10

N	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I	k	α_{95}
6	343	32			5	32
			336	30	30	12

Ages are lower Miocene. Other comments as for Table 1.

below on the data of *Pozzi and Feinberg* [1991] in the Tadjik depression). Third, corresponding inclination anomalies would be expected in other parts of the world, but have not been observed so far.

Other local processes can be invoked to account for an inclination shallowing.

1. Uncertainity on the tilt correction can lead to an inclination anomaly. However, most of our samples were taken from well-bedded rocks on both sides of a fold and fold tests are positive at the 99% level [*Mc Elhinny*, 1964], excluding large errors in bedding measurement or in tilt correction.

2. Large penetrative strain can create an inclination anomaly. Measurement of anisotropy of magnetic susceptibility is one way to test the influence of strain on magnetization [*Hrouda*, 1981]. The anisotropy factor (ratio between the largest and smallest principal values of the anisotropy tensor) measured for some localities of the Tadjik depression is generally lower than 5%, which is too small for a significant deviation of the magnetization [*Nagata*, 1961].

3. Compaction can be responsible for inclination shallowing, especially in clays [*King*, 1955; *Blow and Hamilton*, 1978; *Anson and Kodama*, 1987; *Arason and Levi*, 1990]. For rocks such as redbeds, dominated by quartz grain aggregates, a single study by *Stamatakos et al.* [1989] yielded no significant inclination anomaly. Furthermore, while clays can be strongly compacted (up to 60%), sandstones are probably not liable to noticeable compaction [e.g., *Meade*, 1966].

4. Laboratory experiments and field studies have shown that for detritic magnetization, an important inclination error can be observed [*King and Rees*, 1966; *Tauxe and Kent*, 1984]. However, the magnetization of our samples seems to be carried mostly by hematite pigment of chemical origin. Furthermore, in the Tadjik depression and in the Kirghiz Tien-Shan [*Thomas et al.*, 1993], we sampled several different kind of sediments, ranging from fine-grained to coarse-grained silty sandstones, but all showed comparable inclination anomalies. Thus the 30° inclination anomaly is probably not due to errors of this kind.

The processes invoked above can hardly explain inclination anomalies of 30°, unless all factors contribute in the same way, which is unlikely. At the moment, we do not have any satisfactory solution to this problem, which will be analyzed in greater detail in seperate papers.

This anomaly in inclination does not extend to declination. We will see below that declinations measured in Tertiary rocks lead

Table 5. Paleomagnetic Data for Pulkhakim Locality

Site	Latitude, °N	Longitude, °E	Bedding	n/N	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I	k	α_{95}
PLB	38.091	67.426	213/12	8/8	031	41	20	39	15	15
PLC	38.091	67.426	219/32	5/6	037	42	12	34	22	17
PLD	38.091	67.426	219/32	3/6	197	-37	181	-21	23	26
PLE	38.091	67.426	224/22	7/8	343	54	334	30	36	10
PLF	38.091	67.426	214/26	3/9	002	63	336	44	34	21
PLG	38.091	67.426	214/25	7/8	004	73	329	54	30	11
PL1	38.060	67.475	20/83	6/8	334	-11	359	42	20	15
PL2	38.060	67.475	18/94	5/6	170	21	176	-28	28	15
PL3	38.060	67.475	17/93	5/6	167	38	154	-22	83	8
PL4	38.060	67.475	4/118	8/9	166	56	134	-33	21	12

N	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I	k	α_{95}
10	0	22			3	34
			349	35	16	13

Ages are lower Miocene. Other comments as for Table 1.

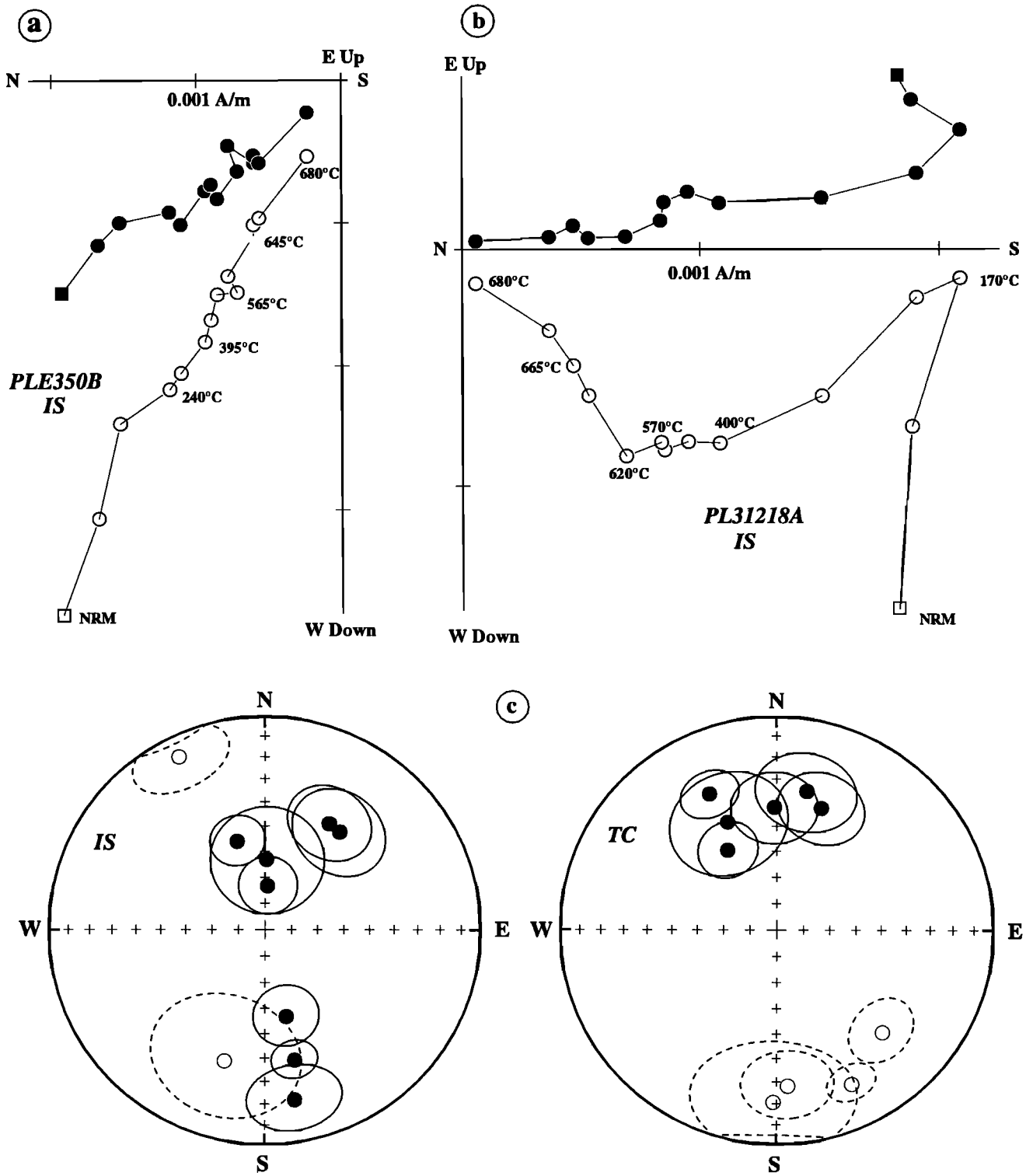


Figure 9. Paleomagnetic data for Pulkhakim locality. Same comments as for Figure 5.

to inferred rotations almost identical in sense and magnitude to those inferred from Cretaceous rocks which show no inclination anomaly [*Bazhenov et al.*, 1994].

Declination

In contrast with inclinations, measured declinations are strongly variable across the Tadjik depression (Figure 11b and Table 7). Compared with the Dekhanabad reference (which has undergone no rotation), two geographic domains of declination can be distinguished (Figure 12).

1. In the eastern part of the depression, Tukaynaron, Kalinabad and South Darvaz localities show large counterclockwise paleomagnetic rotations (46°±14°, 54° and 52°±13°, respectively).

2. In the west, Aksu, Pulkhakim and Pyryagata localities show smaller counterclockwise rotations (17°±16°, 14°±15° and 27°±14°, respectively). Uncertainties are large for Aksu and

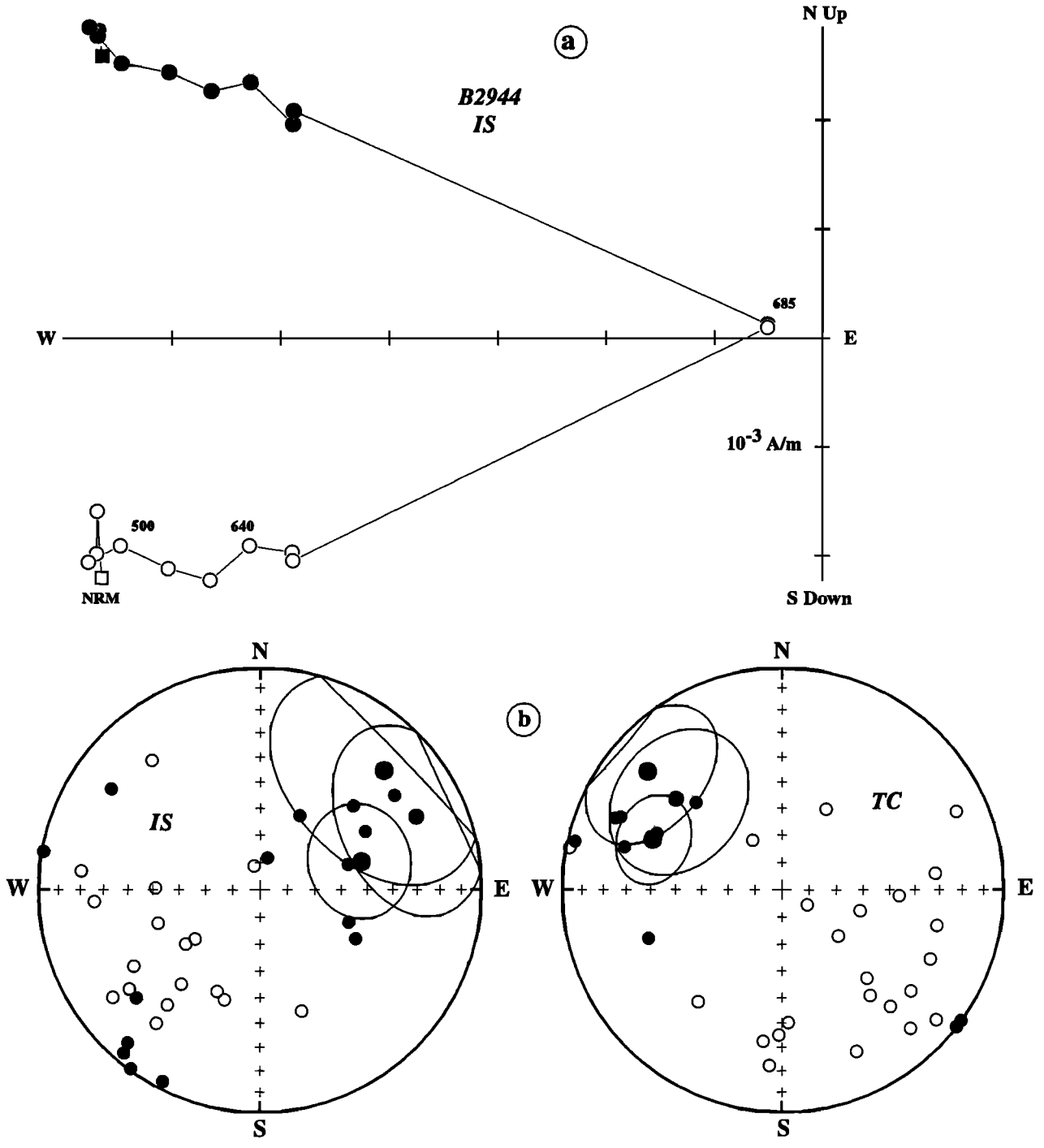


Figure 10. Paleomagnetic data for South Darvaz locality. Same comments as for Figure 5.

Pulkhakim localities. However, at Aksu locality, Cretaceous rocks sampled by *Bazhenov et al.* [1994] show a significant counterclockwise rotation of 15°±5° (Figure 12 and Table 7) with respect to the *Besse and Courtillot* [1991] APWP (Table 7). Thus with two independent references, Cretaceous and Tertiary samples from the same fold have yielded very similar paleomagnetic rotations. We assume that rotations for Tertiary and Cretaceous rocks have the same tectonic origin. If so, the rotation recorded at Pulkhakim may also be significant.

Table 7 shows the amounts of rotation calculated with respect to the APWP of *Besse and Courtillot* [1991] at 30 Ma. At Aksu and South Darvaz localities, this leads to rotations which are larger for Tertiary rocks than for Cretaceous rocks. This rather unexpected result strengthens our choice of Dekhanabad locality as a convenient reference for our study.

Other data

Figure 12 (see also Table 7) shows all available paleomagnetic data on Mesozoic and Cenozoic rocks in the Tadjik depression [*Pozzi and Feinberg*, 1991; *Bazhenov et al.*, 1994; *Chauvin et al.*, 1992; this study]. Soviet data were discarded because of their generally poor quality (due to a small number of sites and

Table 6. Paleomagnetic Data for South Darvaz Locality

Site	Latitude, °N	Longitude, °E	n/N	In Situ D	In Situ I	In Situ k	In Situ α_{95}	Tilt Corrected D	Tilt Corrected I	Tilt Corrected k	Tilt Corrected α_{95}
S1	37.830	68.230	13/19	74	51	5	20	291	37	9	15
S2	37.507	68.031	9/9	46	24	3	38	311	20	6	24
S3	37.715	68.150	6/12	65	24	5	31	310	37	7	26

	n/N	In Situ D	In Situ I	Tilt Corrected D	Tilt Corrected I	k	α_{95}
Site level	3/3	30	34			17	30
				304	32	35	21
Sample level	28/40	62	37			4	16
				312	32	7	11

See text for ages. Other comments as for Table 1.

Table 7. Poles and Associated Reference Paleomagnetic Directions for Eurasia at 30 Ma and 100 Ma and Their Comparison With Available Data in the Tadjik Depression

	Latitude °N	Longitude °E	A95
Besse and Courtillot [1991], Pole (100 Ma)	76.7	197.1	5.0
Besse and Courtillot [1991], Pole (30 Ma)	81.0	132.8	2.7
Van der Voo [1990], Pole (67-97 Ma)	74.0	155.0	
Van der Voo [1990], Pole (37-66 Ma)	78.0	178.0	
[*Chauvin et al.*, 1992], Dekhanabad (lower Miocene)	68.0	239.0	9.0

	Inclination Difference: Versus Besse and Courtillot Pole	Inclination Difference: Versus Dekhanabad Locality Pole	Rotation: Versus Besse and Courtillot APWP	Rotation: Versus Dekhanabad Locality
		Oligo- Miocene, (This Study)		
South Darvaz	+28°±09°	+2°±15°	-59°±10°	-52°±13°
Tukaynaron	+25°±9°	+5°±11°	-54°±12°	-46°±14°
Kalinabad	+29°	-1°	-62°	-54°
Aksu	+27°±12°	+3°±17°	-24°±15°	-17°±16°
Pyryagata	+30°±10°	0°±12°	-35°±12°	-27°±14°
Pulkhakim	+24°±11°	+5°±16°	-21°±13°	-14°±15°
		Upper Cretaceous, Bazhenov et al. [1994]		
South Darvaz	-7°±6°		-51±7°	
Aksu	+1°±5°		-15°±5°	
Derbent	+1°±3°		-05°±5°	
Takob	+4°±6°		-09°±6°	
		Upper Cretaceous, Pozzi and Feinberg [1991]		
Nurek	+7°±8°		-32°±15°	
Guissar	-4°±5°		-05°±7°	
Ragoon	-12°±4°		-13°±8°	

Differences in inclination and declination (rotation) between reference and measured data are calculated according to *Beck* [1980] and *Demarest* [1983]. Clockwise rotations are positive. Positive inclination differences are for inclinations lower than expected. A95 is radius of circle of confidence at 95% probability level.

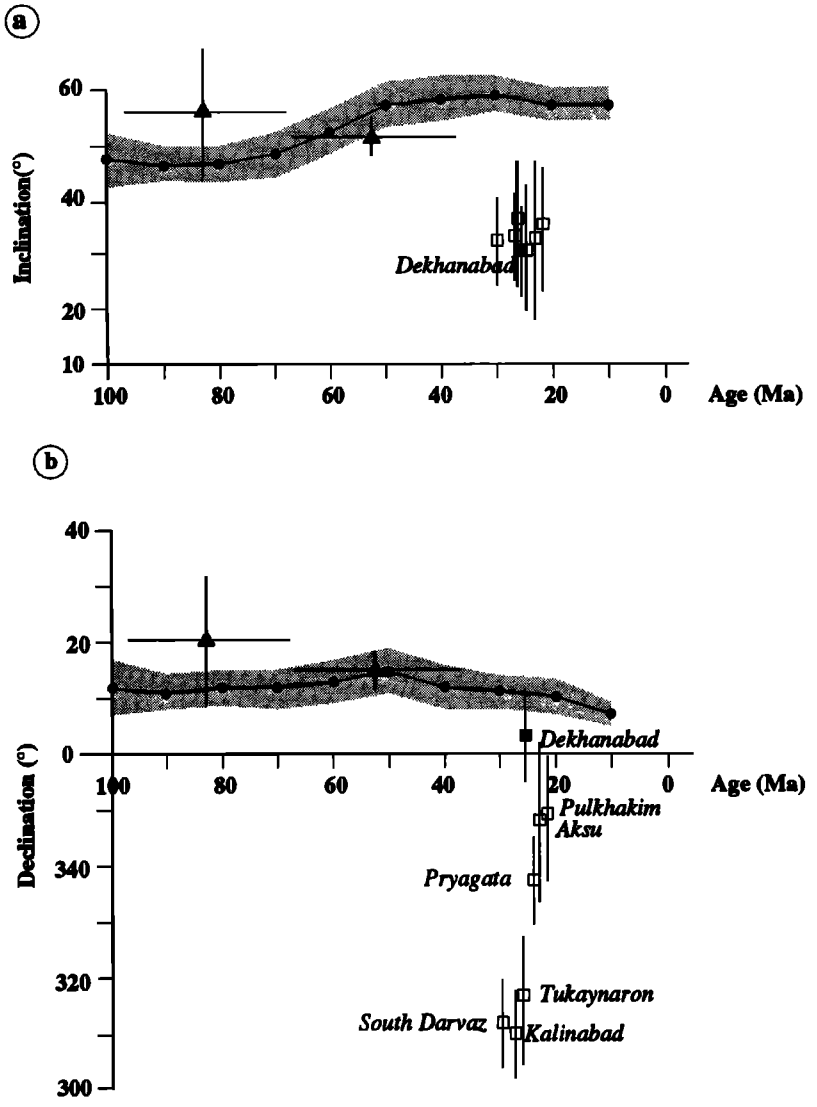


Figure 11. Comparison of our available paleomagnetic data with apparent polar wander paths (APWPs). Graphs show paleomagnetic (a) inclination or (b) declination as a function of age. APWP of *Besse and Courtillot* [1991] is represented by segmented curve joining mean values (solid circles) between 95% confidence limits (shaded strips). APWP of *Van der Voo* [1990] is represented by two data points (solid triangles) with error bars. Paths were calculated for a point at 38°N, 68°E. For our data, inclinations are consistently about 30° shallower than expected, whereas declinations vary widely but are all counterclockwise of expected values. Reference values (solid square) are provided by Dekhanabad locality [*Chauvin et al.*, 1992] near the Turan platform.

incomplete progressive demagnetization). *Pozzi and Feinberg* [1991] studied Cretaceous sediments from three localities in the northwestern part of the depression. They interpreted the mean direction as a Tertiary overprint linked to the Alpine-Himalayan orogenesis. However, *Bazhenov et al.* [1994] studied equivalent formations and showed that the overprint is more probably Upper Cretaceous. We therefore chose an Upper Cretaceous reference to reinterpret data of *Pozzi and Feinberg* [1991]. Corresponding directions must be interpreted with care, because of the small number of samples per locality.

The Cretaceous data confirm and complete the Tertiary data, especially in the northern and western parts of the study area. In the Ghissar and SW Ghissar ranges, small to insignificant paleomagnetic rotations are observed (5°±5° for Ghissar, 9°±6° for Takob, 5°±5° for the SW Ghissar).

Tectonic Interpretation

The major inferences from the declination data are as follows:

1. Tertiary and Cretaceous rotations are very similar. Thus we infer that no significant rotations occurred between the Upper Cretaceous and the lower Miocene.

2. Significant counterclockwise rotations occurred within the Tadjik depression. Rotations were larger in the eastern part than in the western part.

3. Little or no rotation occurred in the SW Ghissar ranges and in the Ghissar range.

Furthermore, tectonic data suggest the Tadjik depression as a lozenge-shaped basin bounded by thrust-wrench zones (Figure 13).

In their interpretation of the rotation pattern observed in

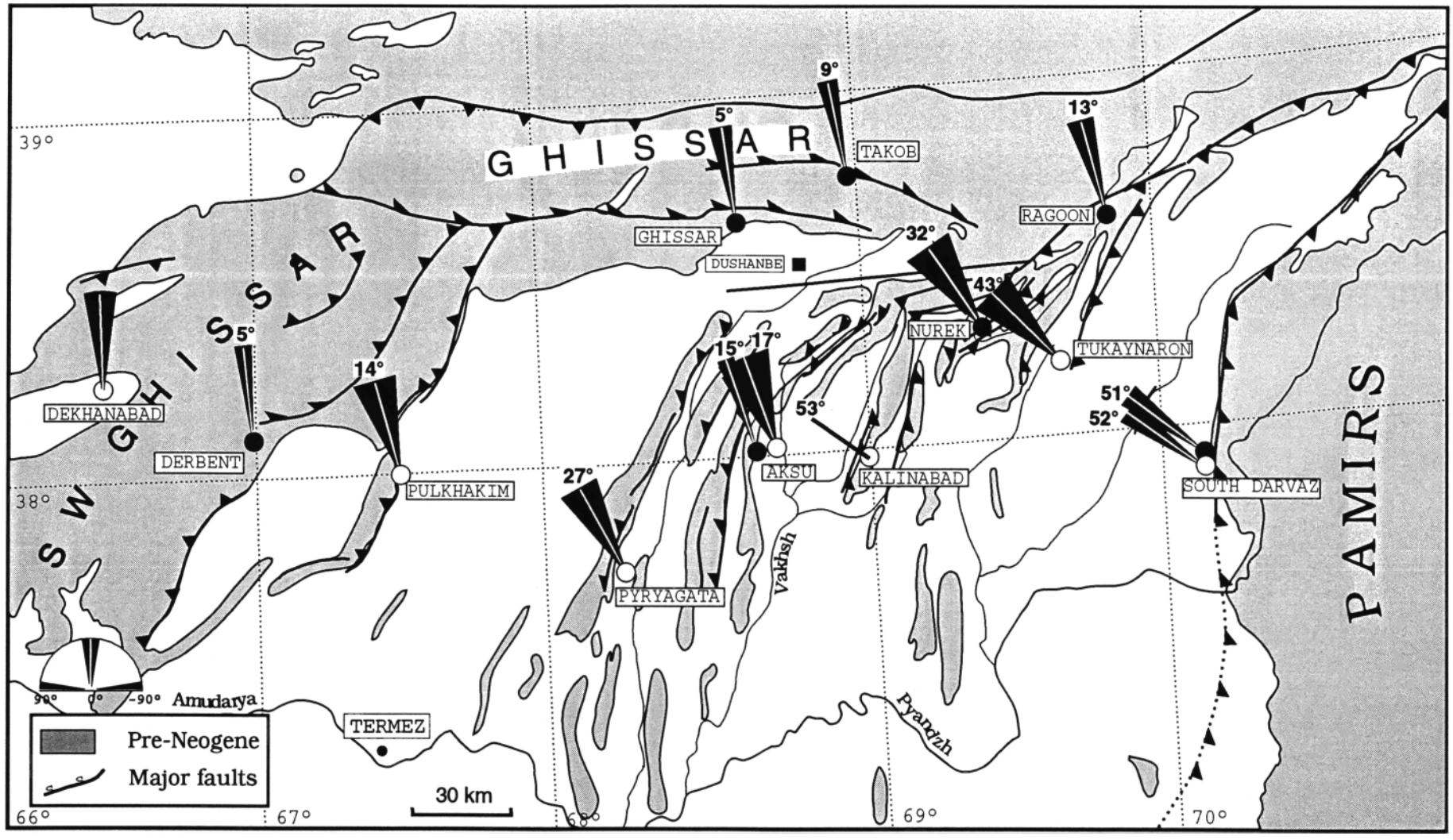


Figure 12. Compilation of paleomagnetic rotations for the Tadjik depression. Rotations are calculated using a *Besse and Courtillot* [1991] pole at 100 Ma for Cretaceous data (solid circles), and using Dekhanabad locality [*Chauvin et al.*, 1992] for Tertiary data (open circles). Rotations (absolute values) and attached uncertainties are shown with respect to a N-S reference direction. Rotations are all counterclockwise, larger in the eastern part of the depression than in the western part of the depression. Little or no rotation has been measured in the Ghissar and SW Ghissar ranges.

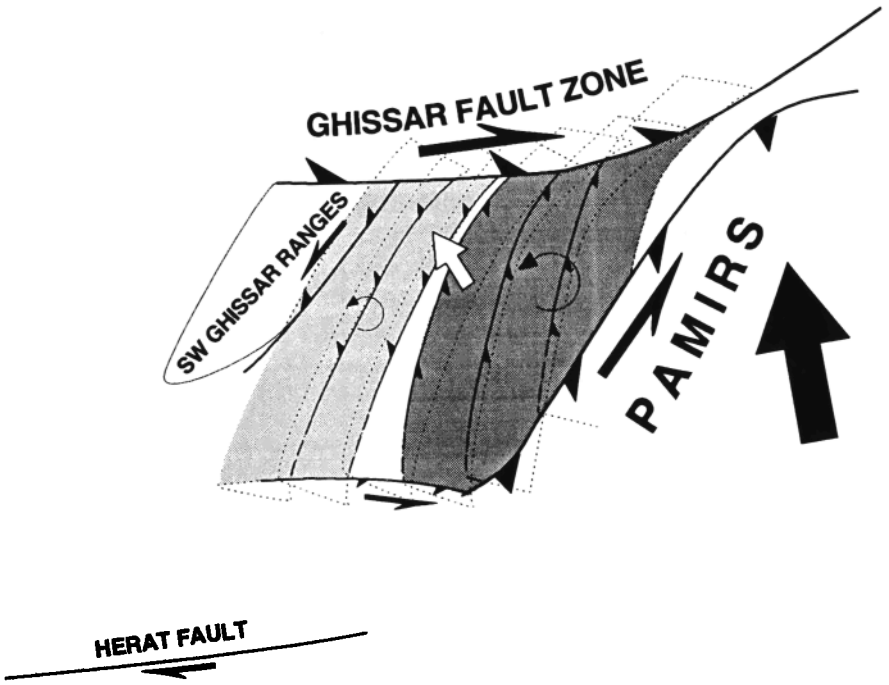


Figure 13. Inferred Cenozoic kinematics of the Tadjik depression. solid symbols indicate motion of boundary blocks relative to the depression itself. Thus the Pamirs move NNW (large solid arrow), this being decomposed at the edge of the depression on to components of overthrusting to the NW (triangles pointing in direction of underthrusting) and left-lateral strike slip (long halfarrow). The Ghissar block overthrusts the depression southwards and moves right-laterally. The SW Ghissar ranges overthrust to the ESE and move left -laterally. The depression itself undergoes oblique internal thrusting (small solid triangles). Underthrust edges of blocks are indicated schematically (dashed lines). Rotations are large in eastern part of depression (dark grey); smaller in western part (light grey). Central domain (open) is a ramp basin. Rotation of the Tadjik depression is accompanied by some north-westward extrusion (open arrow).

intermontane basins of the Western Tien-Shan, *Thomas et al.* [1993] and *Bazhenov* [1993] have suggested that regional kinematics were strongly associated with the indentation of the Pamirs into Asia. Current plate motions indicate a NNW to N-S movement of the Pamirs [*Minster and Jordan*, 1978; *De Mets et al.*, 1990]. In the Tadjik depression, structural analysis [*Thomas et al.*, 1994] has revealed that thrusting is combined with a regional component of sinistral wrenching, synthetic to the relative motion of the Pamirs with respect to the Turan Platform. Paleomagnetic data presented above show that this sinistral wrenching is accompanied by counterclockwise rotations about vertical axes. Both counterclockwise rotations and N-S to NE-SW sinistral wrenching fit the regional model of Tertiary kinematics suggested by *Cobbold and Davy* [1988] for the area.

The Tadjik depression has two paleomagnetic domains, with large rotations near the Pamirs, and smaller rotations near the SW Ghissar ranges (Figure 13). These also appear to be major structural domains of the depression (Figures 1b and 13): the eastern part of the depression shows westward directed overthrusts; whereas the western part shows eastward directed overthrusts. A pop-down (full ramp) basin occurs between these two domains. The occurrence of opposite senses of strike-slip components along the northern and southern boundaries of the depression (Figure 13) suggests that rotation of the depression was accompanied by some northwestward extrusion. Thick-skinned deformation occurs at the margins of the Tadjik depression (Ghissar, SW Ghissar, Pamirs). The deformation inside the depression, especially in its central part, partly involves basement (Figure 2b). From this, we infer that block rotations in the brittle crust have probably occurred. The importance of the Upper Jurassic décollement level remains, however, unknown.

In the SW Ghissar ranges, no significant paleomagnetic rotations have been found. Except in the northermost part of these ranges, deformation decreases gradually toward the stable Turan platform. We therefore infer that the SW Ghissar ranges remained relatively fixed during the deformation, with respect to the Tadjik depression and the Turan platform. Thrusts involving basement are much more developed in the northern part of the SW Ghissar ranges than in their southern part (Figure 2a). Topographic maps show that the uplifted area has a triangular shape, with the apex to the south. These two features suggest that the amount of horizontal convergence between the Tadjik depression and the Turan platform decreases southward (Figure 13). This in turn indicates that the mean pole of counterclockwise rotation of the depression is located somewhere in its southwestern part.

North of the Tadjik depression, the E-W striking Ghissar ranges form a major uplifted fault zone which curves counterclockwise, from NW-SE in the vicinity of the stable platform, to WSW-ENE north of the Pamirs (see Figure 2). This change in orientation coincides with increasing regional deformation. Counterclockwise rotations have occurred in areas north and south of the Ghissar range and neighboring mountain belts, in the Fergana basin and in the Tadjik depression, respectively. These features suggest that substantial counterclockwise rotations could have affected the Ghissar range.

In contrast, our paleomagnetic study reveals minor to negligible counterclockwise rotations of this domain. Two end-members hypotheses can be invoked. (1) Regional counterclockwise rotations have been partly compensated by local clockwise rotations. This could for instance have been achieved by rotation of second-order antithetic dominoes within the Ghissar dextral wrench zone. (2) The Ghissar range is a relatively fixed domain, uplifted between two rotated domains. This hypothesis implies substantial amounts of underthrusting of blocks north and south of the ranges. Indeed, structural data show that substantial thrusting has accompanied dextral wrenching throughout the Ghissar range and at the northern boundary of the Tadjik depression. Other features are in favor of limited rotation of the Ghissar range compared with the Tadjik depression. First, maps [*Ministry of Geology of the USSR and the Tadjik Academy of Sciences*, 1984] show that the Ghissar range is associated with a single set of numerous major thrust-wrench faults parallel to the belt and probably of lithospheric scale; but they do not show evidence for well-developed sets of antithetic dominoes. Second, the major crustal thickening expected in this area is likely to limit its ability to rotate, compared with the less thickened crust beneath the Tadjik depression. Third, faults and fold axes within the Tadjik depression strike about N-S in the center of the depression and curve to become at a low angle to the strike of the Ghissar range in its vicinity (Figure 2). This curvature is compatible with dextral wrenching along the Ghissar range; but it is also compatible with a jump in the amount of rotation between blocks of the Tadjik depression and the Ghissar range. Sandbox experiments [*Gapais et al.*, 1991] have indeed shown that bending of faults can accommodate jumps in rigid rotation across incoherent boundaries between domains of conjugate fault sets.

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